

Throughput Scaling of a GPU Speech Transcription and Diarization Service: Threads Fail, Processes Win, and a Gateway Ties Them Together

Gregorio Toscano*

The Catholic University of America
toscanopulido@cua.edu

Grace Connors

The Catholic University of America
connorsg@cua.edu

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Abstract

Speaker-attributed speech transcription combines automatic speech recognition (ASR), forced alignment, and speaker diarization into a single pipeline. Such pipelines are widely deployed as GPU services, yet their concurrency behaviour is poorly characterised: practitioners routinely assume that a GPU-resident model is GPU-bound and that adding worker threads will raise throughput. We show this assumption is wrong for the popular *WhisperX* pipeline (Whisper **large-v3** + wav2vec2 alignment + pyannote diarization). Through controlled measurements on an NVIDIA RTX PRO 6000 GPU we find that (i) a single transcription stream leaves the GPU only 40–44% utilised because alignment and diarization are CPU- and Python-bound, (ii) allowing a second worker *thread* to run concurrently within one process *reduces* throughput by 28% due to contention on the CPython Global Interpreter Lock (GIL) and the GPU, while (iii) replicating the service as independent *processes* yields near-ideal speedup initially, followed by diminishing returns ($1.50\times$ at three replicas for the short workload and $1.70\times$ at four for the long workload; 73–81% utilisation). We then introduce a lightweight nginx *gateway* that presents a single endpoint and round-robins requests across replicas; because each replica’s job registry is process-local, we route job status requests back to the owning replica by encoding the replica identity in the job identifier. Across five trials, the gateway shows no throughput penalty: its mean is 1–2% above direct routing, within the observed execution variability. Finally we give a remainder-aware wave model for the wall-clock time of a batch of N files on R replicas. Three end-to-end trials show that the synchronized-wave estimate is conservative for continuously queued work, overpredicting the mean wall time by 4–7%. Our results give practitioners a concrete recipe for sizing single-GPU speech services and a cautionary tale about thread-level parallelism in GIL-bound GPU pipelines.

1 Introduction

Automatic transcription of long-form, multi-speaker audio—meetings, hearings, interviews, podcasts—has become a commodity workload, driven by the accuracy of the Whisper family of models [1] and by open pipelines such as WhisperX [2] that add word-level timestamps and speaker labels. A common deployment pattern is a GPU-resident service: a long-running process that loads the models once and serves transcription requests over HTTP.

When such a service must handle several recordings concurrently, the operator faces a sizing question: *how many simultaneous transcriptions can one GPU sustain, and what wall-clock time should a batch of N files take?* The folk answer is that a GPU model is GPU-bound, so one should raise an intra-process worker count until the GPU saturates. We show that this reasoning fails for speaker-attributed transcription, for two compounding reasons. First, only part of the pipeline runs on the GPU: forced alignment and, especially, neural speaker diarization spend substantial time in Python and on the CPU, so a single stream leaves the GPU well under half utilised. Second, because that work runs in Python, additional *threads* cannot execute

*Corresponding author.

it in parallel—the CPython Global Interpreter Lock (GIL) [10] serialises Python bytecode—so thread-level workers contend on the GIL and on the GPU and make throughput *worse*.

We make the following contributions:

- A measurement study (§5) quantifying the GIL/CPU bottleneck in the WhisperX pipeline and showing that thread-level parallelism regresses throughput while process-level replication scales it.
- A single-GPU replica architecture with an nginx gateway (§3) that gives clients one endpoint. We identify and solve the *process-local job registry* problem by encoding replica identity in the job identifier and routing status requests accordingly.
- An evaluated, remainder-aware wave model (§5.5) that conservatively estimates homogeneous batch completion time from concurrency-specific service times.

All experiments use a single GPU and off-the-shelf components; our aim is a reproducible recipe rather than a new model or kernel. The complete implementation—service, multi-GPU launcher, gateway, and benchmark suite—is open source.¹

2 Background

The WhisperX pipeline. For each request the service runs four stages: (1) *ASR*: Whisper `large-v3` [1] executed through faster-whisper [9] / CTranslate2 [8], producing text with coarse segment timestamps; (2) *alignment*: a wav2vec2 phoneme model [3] forces word-level timestamps; (3) *diarization*: the pyannote `speaker-diarization-3.1` pipeline [4, 5, 6] segments the audio by speaker; (4) *assignment*: words are attributed to speakers and relabelled. Stages 1 and 2 are dominated by GPU tensor operations, but stage 3 performs significant CPU-side clustering and stages 2–4 involve substantial Python-level orchestration.

The GIL. CPython executes at most one thread of Python bytecode at a time, guarded by the GIL [10]. Native extensions (PyTorch kernels, CTranslate2) release the GIL while running C++/CUDA code, so a single Python thread *can* keep the GPU busy. However, any pipeline phase that stays in Python—graph manipulation, clustering bookkeeping, segment assignment, audio I/O—holds the GIL, and two Python threads cannot run those phases simultaneously.

Thread-safety. An orthogonal constraint: the pyannote 3.x diarization pipeline is not thread-safe, so a shared model instance cannot be driven from multiple threads regardless of the GIL. Any concurrency scheme must therefore give each worker its own diarization model.

3 System Architecture

3.1 Baseline service

The service is a FastAPI/Uvicorn application in a Docker container. It exposes an asynchronous submit endpoint that enqueues a job and returns a job identifier, plus polling endpoints for status and results. A configurable number of worker tasks (`WORKERS`) drain the queue; each worker owns a private *model set* (ASR, per-language alignment, diarization) so that diarization is never shared across threads. Results are also written to a shared volume as files, so completed transcripts survive process restarts.

3.2 Thread-level parallelism (the tempting wrong answer)

Setting `WORKERS > 1` spawns multiple worker threads in the one process, each with its own model set, and runs each job’s blocking pipeline in a thread pool. This is the natural first attempt at concurrency. §5 shows it does not work: the Python-bound phases serialise on the GIL and the threads additionally contend for the GPU.

¹<https://github.com/gtoscano/Parascribe>

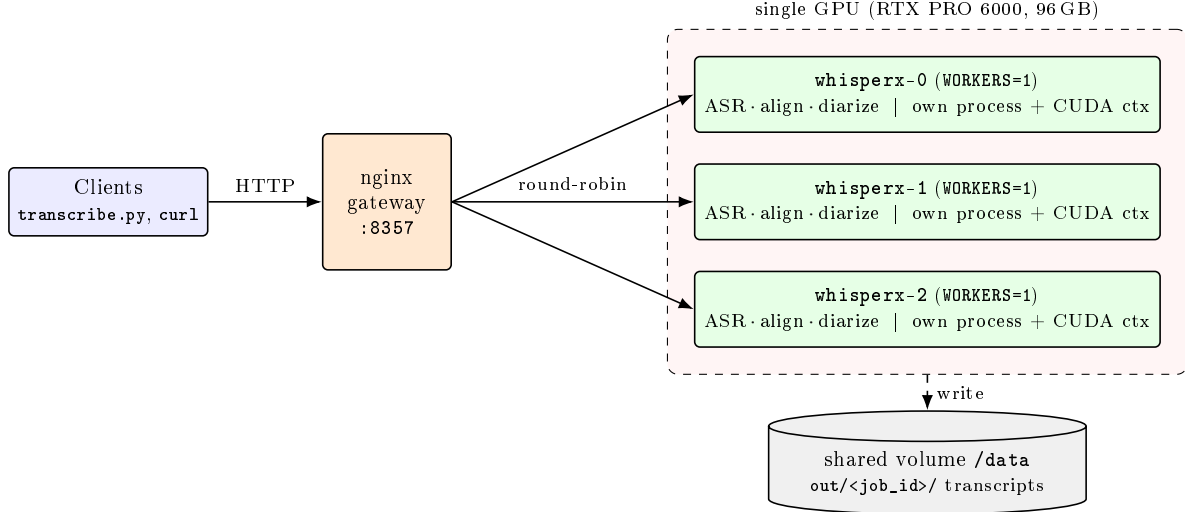


Figure 1: System architecture. Clients issue HTTP to a single nginx gateway (:8357). The gateway round-robins POST /v1/transcribe across the R process replicas and routes each GET /v1/jobs/ i -... back to the owning replica i by the numeric job-id prefix, so the process-local job registries need no shared state. Each replica is an independent OS process—its own Python interpreter (hence its own GIL) and CUDA context—running WORKERS=1 with a private ASR/align/diarize model set; all R replicas share one GPU. Finished transcripts are written to a shared /data volume.

3.3 Process-level replication

Real parallelism requires separate interpreters. We run R identical container *replicas* of the service, each with WORKERS = 1, all pinned to the same GPU. Each replica has its own Python process (own GIL) and its own CUDA context, so up to R jobs execute genuinely concurrently, bounded only by GPU compute and memory. On our hardware a replica’s resident footprint is ≈ 13 GB, so memory is not the binding constraint on a 96GB GPU; compute is.

3.4 The gateway and the process-local registry problem

Exposing R replicas to clients as R URLs pushes load-balancing into every client. We instead front the replicas with an nginx gateway that presents a single endpoint and round-robins requests (Figure 1). One subtlety makes a naïve proxy incorrect: each replica’s job registry is *in-memory and process-local*. A job submitted to replica A is unknown to replica B , so a status poll that a plain round-robin proxy sends to B returns 404.

We solve this without shared state by encoding the owning replica in the job identifier. Replica i mints identifiers of the form i -{token}. The gateway round-robins only the submit endpoint; for job-scoped requests it extracts the numeric prefix and routes to that replica (Listing 1). Because the transcript files are also written to a shared volume, results remain retrievable even across restarts.

```

resolver 127.0.0.11;                                # Docker embedded DNS
upstream pool { server r0:8000; server r1:8000; server r2:8000; }

# Job status/results -> the replica named in the id prefix
location ~ ^/v1/jobs/(?<rid>[0-9]+)- {
    set $backend r$rid:8000;
    proxy_pass http://$backend;
}

# Submit + health + models -> round-robin the pool
location / { proxy_pass http://pool; }
  
```

Table 1: Workloads.

Tag	Source	Duration	Speakers
W-short	AMI meeting	477.9 s ($\approx$ 8 min)	4
W-long	Oyez argument	2400 s (40 min)	8

Listing 1: nginx routing: round-robin submits, prefix-route job lookups.

Two operational notes proved important in practice: a single nginx worker process is preferable because per-worker round-robin counters otherwise skew the distribution on small samples; and the gateway must wait for replicas to become healthy before serving, or its failover logic transiently concentrates load on the first ready replica.

4 Experimental Setup

Hardware. A workstation with two NVIDIA RTX PRO 6000 (Blackwell) GPUs, 97 887 MiB ($\approx$ 96 GiB) each. The transcription service is pinned to one GPU; the other hosts an unrelated model and is not used by our experiments.

Software. Whisper `large-v3` via faster-whisper [9] / CTranslate2 4.8.0 [8], float16, batch size 16; wav2vec2 English alignment [3]; the pyannote `speaker-diarization-3.1` pipeline [5, 6] on PyTorch [7]; FastAPI + Uvicorn; Docker with the NVIDIA runtime; nginx (Alpine) as the gateway.

Workloads. Two representative recordings (Table 1): a short 8-minute meeting excerpt (AMI corpus [11], 4 speakers) and a long 40-minute oral argument (Oyez, 8 speakers). The short workload isolates per-request overheads and lets us sweep concurrency cheaply; the long workload reflects realistic batch jobs.

Methodology. A load generator submits c identical jobs concurrently and records: wall-clock time for the batch; per-job service time (server-side running $\rightarrow$ finished); GPU memory, utilisation and power draw sampled at 2 Hz via `nvidia-smi`; host RAM; and host CPU occupancy derived from `/proc/stat` deltas. CPU is reported in *cores busy* rather than a percentage, since a percentage is not actionable for provisioning without the core count. Per-replica CPU and resident set size are attributed separately via `docker stats`, because the measurement host also runs unrelated services and a host-wide figure cannot be divided by the replica count. All resource figures are reported as deltas over an idle baseline measured immediately before each sweep.

The service itself is instrumented: each stage of the pipeline (audio load, model preparation, transcription, alignment, diarisation, speaker assignment, serialisation, output write) is timed individually and persisted per job, alongside the queue wait. This separates *preparing* a model from *using* it, a distinction that a single end-to-end duration cannot express and that Section 5 relies on to attribute contention to a specific stage. We report throughput in files min^{-1} and the real-time factor $\text{RTF} = (\text{audio seconds})/(\text{wall seconds per file})$. Models are warmed once per service configuration. We then invoke the load generator independently five times for every thread, process-scaling, and gateway configuration; the two expensive ten-file batch validations are repeated three times. Each invocation records a fresh idle baseline, and multi-level sweeps use a 5-second cooldown between levels. Unless stated otherwise, values are arithmetic mean and sample standard deviation, written *mean* (*SD*); speedups are ratios of repeated means. Where an interval is given it is a two-sided 95% confidence interval computed with Student’s t ; at five repetitions the t multiplier is 2.776, and using the normal approximation would understate the interval by roughly 40%. All retained trials had zero failed jobs. The raw per-trial CSV files accompany the paper in `measurements/repetitions/`; the stage-level and resource measurements of Section 5 are in `measurements/ami_2rep_1gpu_14r5/`, each run carrying a provenance record naming the code revision, GPU driver and container image that produced it.

Table 2: Thread-level parallelism on W-short (single process, `WORKERS=2`). Activating both worker threads does not improve throughput; four submitted jobs execute in two waves. Entries are mean (sample SD), $n = 5$.

In-flight jobs	Throughput (files min ⁻¹)	GPU util. (%)	Per-job (s)
1	5.94 (< 0.01)	40.4 (2.1)	8.9 (0.0)
2	4.25 (< 0.01)	31.1 (0.8)	27.0 (0.3)
4	4.25 (< 0.01)	32.0 (1.3)	27.3 (0.2)

Table 3: Process-level replication. Throughput and GPU utilisation are mean (sample SD), $n = 5$; speedup is the ratio of throughput means.

Replicas	Workload	Throughput (files min ⁻¹)	GPU util. (%)	Speedup
1	W-short	5.94 (< 0.01)	43.4 (2.1)	1.00
2	W-short	8.49 (< 0.01)	71.3 (1.5)	1.43
3	W-short	8.89 (0.02)	73.5 (1.6)	1.50
1	W-long	0.99 (< 0.01)	43.8 (0.4)	1.00
3	W-long	1.54 (< 0.01)	74.0 (0.6)	1.56
4	W-long	1.68 (0.01)	80.6 (0.6)	1.70

5 Results

5.1 A single stream leaves the GPU idle

On W-short a single in-flight request sustains $5.94 (< 0.01)$ files min⁻¹ ($47.3 \times \text{RT}$) at 43.4 (2.1)% GPU utilisation; on W-long, $0.99 (< 0.01)$ files min⁻¹ ($39.6 \times \text{RT}$) at 43.8 (0.4)%. Nearly half the GPU is idle because alignment and diarization are not GPU-bound. This is the headroom that a concurrency scheme must convert into throughput.

5.2 Threads make it worse

Table 2 holds the service at two worker threads and varies the number of concurrent W-short jobs. Going from one job (one active worker) to two jobs (both workers active) *lowers* mean throughput by 28.5%, from 5.94 to 4.25 files min⁻¹, and triples per-job service time ($8.9 \rightarrow 27.0$ s); mean GPU utilisation *falls* from 40.4% to 31.1%. This is the signature of a GIL/CPU bottleneck: the second thread cannot run the Python-bound phases in parallel, so it adds contention (GIL hand-off, GPU context switching) without adding useful work. Threads are the wrong primitive here. With four submitted jobs, the two workers execute two waves; throughput remains below the single-active-worker baseline.

5.3 Processes scale, up to GPU saturation

Table 3 replaces threads with independent process replicas. Throughput now rises with replica count until the GPU’s compute saturates: $1.50\times$ at three replicas on W-short (73.5% utilisation) and $1.70\times$ at four replicas on W-long. The fourth replica on W-long adds only 9.4% (diminishing returns as utilisation approaches saturation and per-job service time lengthens under contention), so three replicas is the practical sweet spot for a single GPU on this pipeline. Figure 2 plots the speedup against the ideal linear line; Figure 3 shows utilisation approaching saturation.

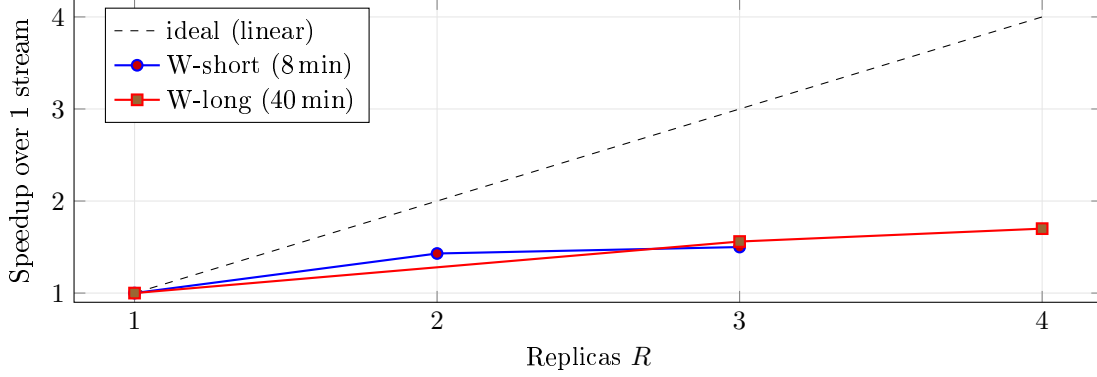


Figure 2: Process replicas deliver real but sub-linear speedup, flattening as the GPU saturates. Error bars are sample SD ($n = 5$) and can be smaller than the markers. Thread-level parallelism (not shown) sits *below* 1.0.

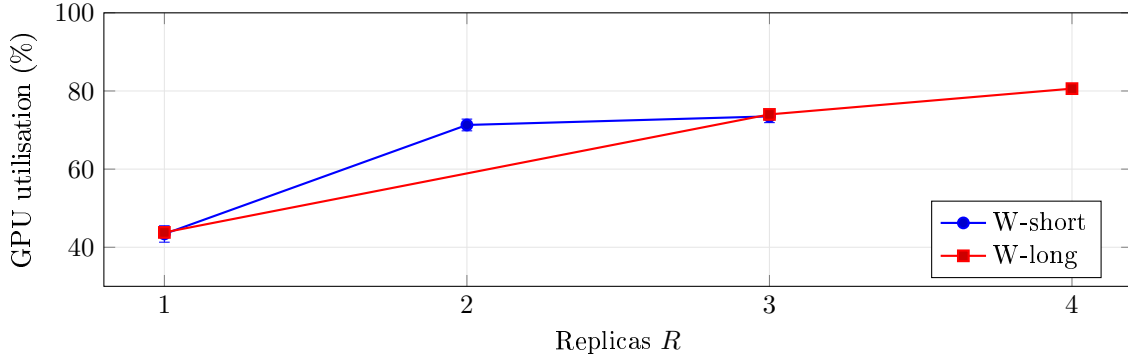


Figure 3: Utilisation climbs from $<50\%$ (single stream) toward saturation as replicas are added. Error bars are sample SD ($n = 5$). The remaining gap on W-long reflects heavier CPU-bound diarization for 8 speakers.

5.4 No gateway penalty is observed

Table 4 compares direct client-side round-robin against routing through the nginx gateway. The gateway distributes submissions evenly and routes status polls back to the owning replica correctly. Its repeated mean is 1.2% higher on W-short and 2.2% higher on W-long. We do not interpret a proxy as accelerating inference; rather, the positive differences expose temporal variation between the unpaired direct and gateway series. The data show no observable throughput penalty, although this descriptive experiment is not an equivalence test.

5.5 A conservative predictive batch model

For a batch of N homogeneous files on R replicas, each serving one job at a time, let

$$q = \lfloor N/R \rfloor, \quad r = N \bmod R,$$

so a synchronized schedule contains q full waves and, when $r > 0$, one partial wave. Let s_j denote the representative server-side service time with j concurrently active replicas; for homogeneous jobs with equal completion times, this is also the wave duration. We estimate s_j by the repeated mean running→finished time. If only the full-wave time s_R is available, a simple estimate is

$$W_{\text{coarse}} \approx \lceil N/R \rceil s_R, \tag{1}$$

Table 4: Gateway vs. direct client-side round-robin, three replicas. Values are throughput mean (sample SD), $n = 5$; difference is $(T_{\text{gateway}} - T_{\text{direct}})/T_{\text{direct}}$.

Workload	Direct (files min ⁻¹)	Gateway (files min ⁻¹)	Difference
W-short	8.89 (0.02)	8.99 (0.43)	+1.2%
W-long	1.54 (< 0.01)	1.57 (0.01)	+2.2%

Table 5: End-to-end batch test through the gateway (Eq. (2)). Predicted and measured times are mean (sample SD); $n = 5$ for input service times and $n = 3$ for batches. Error is $(W_{\text{meas}} - W_{\text{pred}})/W_{\text{pred}}$.

Batch	Predicted	Measured	Error
10× W-short (8 min)	62.5 (0.5) s	60.0 (2.3) s	-4.0%
10× W-long (40 min)	395.0 (2.0) s	370.9 (15.0) s	-6.1%

which conservatively charges a partial wave at the full-wave time when reduced concurrency does not increase service time. When the partial-wave time s_r is known, the remainder-aware estimate is

$$W \approx \begin{cases} q s_R, & r = 0, \\ q s_R + s_r, & r > 0. \end{cases} \quad (2)$$

The earlier single-stream correction is therefore the special case $r = 1$, not a formula for every partial wave.

We test Eq. (2) end-to-end by submitting $N=10$ identical files through the gateway with $R=3$, so $q = 3$ and $r = 1$. For W-short, the repeated means $s_3 = 17.94$ s and $s_1 = 8.70$ s predict 62.5 s; three batches complete in 60.0 (2.3) s. For W-long, $s_3 = 111.76$ s and $s_1 = 59.76$ s predict 395.0 s; the repeated batch mean is 370.9 (15.0) s (Table 5). The prediction uncertainty in the table propagates the sample SDs of s_3 and s_1 under independence; it does not include model-form uncertainty.

The negative signed errors show a systematic conservative bias, larger than the input-time variability. The wave model assumes synchronized boundaries and stationary service times. In a continuously queued batch, however, a replica starts its next job as soon as it finishes; small completion differences then stagger ASR, alignment, and diarization across replicas, smoothing resource use and shortening later service times. Equation (2) is therefore a simple planning estimate, not an exact queueing model. A tighter predictor would need steady-state, stage-aware service measurements or a discrete-event model. For a fixed R , s_R alone still supports the coarse estimate; the refined estimate additionally requires s_r for the batch’s remainder.

5.6 Over-subscription is governed by wave quantisation

The preceding sections vary the number of replicas with offered concurrency held equal to it. A deployment does not get that luxury: the replica count is fixed at provisioning time and the arrival rate is not. We therefore fix a two-replica deployment and vary only the offered concurrency c (Table 6).

Throughput is not monotonic in c . It rises from $c = 1$ to $c = 2$, *falls* at $c = 3$, and returns at $c = 4$ to a value statistically indistinguishable from $c = 2$ (the 95% intervals [8.372, 8.392] and [8.304, 8.416] overlap). A dip that reverses is not contention, which would worsen monotonically.

The explanation is quantisation. With R replicas each running one job at a time, c offered jobs execute in $\lceil c/R \rceil$ waves, and a wave costs what its *widest* occupancy costs. Writing T_k for the wall time of a wave of k concurrent jobs, the batch completes in

$$T(c) = \left\lceil \frac{c}{R} \right\rceil T_R + [c \bmod R > 0] T_{c \bmod R}, \quad (3)$$

so throughput $c/T(c)$ peaks whenever $R \mid c$ and sags otherwise, because a partial final wave occupies the same elapsed time as a full one while completing fewer jobs. With the measured $T_1 = 10.18$ s and $T_2 = 14.31$ s, Equation 3 predicts $T(3) = 24.49$ s and $T(4) = 28.63$ s against observed 24.15 s and 28.71 s — errors of -1.4% and +0.3% with no fitted parameter.

Table 6: Offered concurrency against a *fixed* two-replica deployment (W-short). Throughput is mean (sample SD) with a two-sided 95% t interval, $n = 5$. Efficiency is speedup over $c = 1$ divided by c ; e is the Karp–Flatt experimentally determined serial fraction. Throughput at $c = 4$ recovers to the $c = 2$ value because both are integer multiples of the replica count.

Offered c	Throughput (files min ⁻¹)	95% CI (files min ⁻¹)	Efficiency (%)	Karp–Flatt e
1	5.896 (0.005)	[5.889, 5.903]	99.9	—
2	8.382 (0.008)	[8.372, 8.392]	71.1	0.406
3	7.462 (0.313)	[7.073, 7.851]	45.3	0.603
4	8.360 (0.045)	[8.304, 8.416]	35.6	0.604

Table 7: Server stage means in seconds (share of `pipeline`), fixed two-replica deployment, $n = 5$ per level. Alignment is invariant to concurrency; transcription and diarisation roughly double from $c = 1$ to $c = 2$. Model preparation is 0.00 throughout: the caches are mounted and the run is warmed, so it is excluded by construction.

Stage	$c = 1$	$c = 2$	$c = 3$	$c = 4$
Queue wait	0.00	0.00	4.44	6.61
Audio load	0.14	0.15	0.15	0.13
Transcription	1.88	3.47	2.91	3.46
Alignment	3.02	2.99	3.04	2.99
Diarisation	3.20	6.26	5.26	6.25
Speaker assign.	0.47	0.47	0.47	0.47
Pipeline total	8.72	13.34	11.84	13.30

The practical consequence is that a two-replica deployment serving an odd in-flight count wastes roughly half a wave. Provisioning should therefore match the replica count to the expected concurrency, or admit jobs in multiples of R ; adding replicas that leave $R \nmid c$ can measurably reduce throughput.

Two caveats. First, $c = 3$ is also the only unstable level: its coefficient of variation is 4.2% against $\leq 0.5\%$ elsewhere, and its five observations fall into two clusters (22.4 s once, 24.5–24.7 s four times) rather than scattering about the mean. Equation 3 assumes the partial wave starts only once a full wave has drained, whereas the trailing job can in fact start as soon as the *first* replica frees; the faster cluster is consistent with greater overlap between waves. We report the effect without claiming to have characterised it. Second, the mean of a bimodal level describes a state the system does not occupy, so Table 6 reports dispersion alongside it.

The Karp–Flatt fraction rises from 0.406 at $c = 2$ to ≈ 0.60 at $c = 3$ and $c = 4$, then holds. Read with Equation 3, the step is queueing entering the measurement rather than a larger serial section: the metric attributes the whole of a partial wave’s idle time to serialisation, which is arithmetically correct but conflates two mechanisms. We report it because a rising e distinguishes overhead from an inherently serial section, which speedup alone cannot; we do not read the absolute value as a property of the pipeline.

5.7 Contention is concentrated in ASR and diarisation

Per-stage instrumentation (§4) localises where concurrency is paid for. Table 7 gives the mean duration of each stage at each offered concurrency.

Going from one in-flight job to two, transcription grows $1.88 \rightarrow 3.47$ s and diarisation $3.20 \rightarrow 6.26$ s — both close to a factor of two, meaning the two replicas obtain almost no overlap in these stages. Alignment is flat ($3.02 \rightarrow 2.99$ s): it overlaps essentially perfectly. Speaker assignment and audio load are invariant and negligible.

This is the mechanism behind the 71.1% efficiency at $c = 2$. Two of the three substantial stages serialise

Table 8: Marginal cost of one concurrent job and the resulting ceiling on this host (12 vCPU, 165.1 GiB RAM, one 97887 MiB GPU). GPU memory is *incremental*: the idle baseline already holds 18 625 MiB of resident model weights.

Resource	Per concurrent job	Host ceiling (jobs)
CPU	2.39 cores	≈ 5
GPU memory	3760 MiB	≈ 26
Host RAM	0.09 GiB	≈ 1833

on GPU compute while the third does not, so the achievable speedup is bounded well below linear no matter how many replicas are added to one GPU. It also identifies diarisation — 47% of pipeline time at $c = 2$, the single largest component — as the highest-value target for optimisation, ahead of the ASR pass that dominates the popular understanding of transcription cost.

5.8 Provisioning: the host CPU binds before the GPU

Table 8 reports the marginal resource cost of one concurrent job, measured as a delta over an idle baseline taken immediately before the sweep.

The binding resource is the host CPU, not the GPU. Twelve vCPUs are exhausted at roughly five concurrent jobs, while the 96 GiB GPU would accommodate about twenty-six. For a pipeline whose cost is usually discussed in terms of GPU capability, the accelerator is the abundant resource here and the host is scarce: a deployment that provisions by GPU memory will be CPU-starved long before the card is full.

Throughput saturates earlier still, at $c = 2$ (§5.6), so neither ceiling is reached in practice on this pipeline — the operative limit is GPU compute contention. The ceilings nonetheless bound what additional GPUs can buy: at 2.39 cores per job, twelve vCPUs cannot feed many more concurrent jobs regardless of how many accelerators are added, so scaling this deployment across GPUs requires scaling host cores with them.

Energy is modest: 213 W mean draw at the $c = 2$ operating point, or ≈ 3 Wh per hour of audio transcribed.

6 Discussion

When are threads acceptable? Thread-level workers would help only if the pipeline released the GIL for (almost) its entire duration. A pure-GPU model with a thin Python wrapper can approach that; a pipeline with neural diarization and Python-side clustering does not. The diagnostic is simple and general: if throughput fails to rise—or utilisation falls—as intra-process concurrency increases, the workload is GIL/CPU-bound and needs processes, not threads.

How many replicas? The ceiling is set by GPU compute, not memory: each replica added utilisation until ≈ 73 –81%, after which per-job service time lengthened and returns diminished. Three replicas maximised useful throughput on our single GPU. The heavier, 8-speaker W-long workload saturated at slightly lower utilisation, consistent with a larger CPU-bound diarization fraction. Scaling beyond $\approx 1.7\times$ requires additional GPUs, over which the same replica-plus-gateway design extends directly.

Extrapolation to multiple GPUs. The architecture scales across GPUs without modification. Each replica is pinned to a GPU (via `CUDA_VISIBLE_DEVICES`), and the single gateway round-robins across *all* replicas irrespective of which GPU hosts them; the job-id prefix routing (§3.4) is unchanged. With G GPUs and k replicas per GPU the deployment is $R = Gk$ replicas behind one endpoint. Crucially the workload is *embarrassingly parallel*: each job runs entirely on one replica and one GPU, with no model sharding and no inter-GPU communication, so the GPUs do not contend and throughput scales linearly,

$$T(G) \approx G \cdot T_1, \tag{4}$$

Table 9: Extrapolation to four GPUs at $k = 3$ replicas per GPU. Single-GPU columns are measured; four-GPU columns assume the linear scaling of Eq. (4) across independent GPUs (subject to the host-CPU caveat).

	1 GPU (meas.)	4 GPUs (extrap.)
Replicas $R = Gk$	3	12
Throughput, W-short (files min ⁻¹)	8.89	35.6
Throughput, W-long (files min ⁻¹)	1.54	6.2
W-long recordings / hour	92	370
10×W-short batch (wall)	60.0 (2.3) s	≈ 18 s
10×W-long batch (wall)	370.9 (15.0) s	≈ 112 s
Waves for $N=10$	4	1

where T_1 is the measured single-GPU sustained throughput. For G GPUs with k replicas each, the batch model uses $R = Gk$ in Eqs. (1)–(2); adding GPUs reduces the number of full waves, while the last wave is priced according to the concurrency actually assigned to each GPU.

Table 9 extrapolates to $G = 4$ GPUs at the measured sweet spot $k = 3$ ($R = 12$ replicas). A batch of ten 40-minute recordings falls from four nominal waves (370.9 (15.0) s measured) on one GPU to a single wave (≈ 112 s) on four, and sustained long-file throughput rises from 92 to ≈ 370 recordings per hour. The one caveat is that linear scaling holds only while the *shared* resources keep pace: the gateway (CPU-light) and the `/data` volume (disk I/O, negligible beside transcription) are not the concern, but host CPU is. Because diarization is CPU-bound, each replica consumes host cores, so $R = Gk$ concurrent replicas demand proportional host CPU; on a machine that cannot feed all Gk replicas the CPU—not the GPUs—becomes the ceiling, and k should be lowered accordingly. GPU memory is not a constraint (≈ 13 GB per replica).

Threats to validity. We study one GPU, one pipeline, and two recordings; absolute numbers will differ elsewhere. Repeating the same recording quantifies execution variability, not variability across a population of audio, languages, speaker counts, or acoustic conditions. Five trials per primary configuration and three per batch are enough to expose unstable one-off values but too few for strong distributional assumptions or an equivalence test. An unrelated model on the other GPU shares host CPU, memory, and I/O resources; we leave that realistic background service in place and report within-session sample SD rather than claiming a noise-free machine. Finally, the batch model assumes homogeneous files, stationary concurrency-specific service times, and synchronized waves. The repeated batches expose a 4–7% conservative bias when queued jobs desynchronize. Heterogeneous batches require per-file, stage-aware estimates or a queueing simulation rather than the simple wave sum.

7 Related Work

Whisper [1] and WhisperX [2] established accurate long-form, time-aligned transcription; pyannote provides the diarization stage [4, 5, 6]. Efficient inference backends such as CTranslate2 [8] accelerate the ASR step but do not address pipeline-level concurrency. The limitations of the CPython GIL for multi-threaded compute are long known [10]; recent work on removing it (PEP 703) targets exactly the class of Python-bound parallelism we observe, and would in principle let the thread-level design compete with replication. Our contribution is not a new model or runtime but an empirical characterisation of an inference *service*’s concurrency, the identification of the process-local registry problem for horizontally replicated stateful inference, and an empirically evaluated sizing model—practical guidance that, to our knowledge, is not documented for this now-ubiquitous pipeline.

8 Conclusion

Speaker-attributed transcription is not the GPU-bound workload it appears to be. On a single GPU we showed that a lone stream wastes half the device, that thread-level workers regress throughput because the pipeline is GIL/CPU-bound, and that process-level replication behind a lightweight gateway recovers the idle capacity—1.50× at three replicas for W-short and 1.70× at four for W-long—with no observed proxy penalty. A remainder-aware wave model provides a conservative batch estimate, overpredicting the repeated ten-file means by 4–7%. The recipe is deliberately mundane: replicate the process, encode replica identity in the job id, round-robin at a gateway, and stop adding replicas when the GPU saturates. Because jobs never cross a GPU, the same design scales out linearly—an estimated 4× on four GPUs (Table 9)—until host CPU, not the GPUs, becomes the bound. We hope it saves other practitioners the detour through thread-level parallelism that motivated this study.

A Reproducing the Measurements

All experiments are driven by the cluster manager and load generator in the accompanying repository (<https://github.com/gtoscano/Parascribe>). The cluster manager, `whisperx_cluster.py`, manages the replicas and gateway. The load generator, `benchmark_transcribe.py`, submits `--levels` concurrent jobs per level, warms the models with `--warmup`, samples `nvidia-smi` at 2 Hz, and writes a CSV whose throughput and utilisation columns feed the tables. Each invocation below is annotated with the result it produces. After one warm-up per service configuration, we run each thread, process, and gateway command five times and each ten-file batch command three times, changing only the `--out` suffix (for example, `measurements/repetitions/process_short_r1.csv`). Our deployment places the nginx gateway on port 8357 and the direct replica ports at 8360–8362 (8363 for a fourth replica); adjust paths and ports for your setup.

A.1 Cluster setup

```
# Pre-download the weights. Only the libraries are baked into the image, so
# without this the first request of a fresh deployment pays the download.
python3 fetch_models.py
python3 fetch_models.py --check # exits non-zero if anything is missing

# Build the image and start k replicas on one GPU behind the nginx gateway
REPLICAS=3 python3 whisperx_cluster.py up

# Multiple GPUs (Table 6): k replicas PER GPU, interleaved across --gpus.
# 3 x 4 = 12 replicas behind the same single gateway endpoint:
python3 whisperx_cluster.py up --replicas 3 --gpus 0,1,2,3

# Print the direct per-replica URLs (used by the pool experiments)
python3 whisperx_cluster.py urls --direct
# -> http://localhost:8360,http://localhost:8361,http://localhost:8362
```

A.2 Thread-level parallelism (Table 2)

This experiment uses the single-container service with `WORKERS=2` fixed (not the replica cluster). The load generator varies offered concurrency from one to four jobs: at level 1 only one worker is active, at level 2 both worker threads are active, and at level 4 the two workers process the jobs in two waves.

```
# Set '- WORKERS=2' in docker-compose.yml, then:
docker compose -p whisperx-cluster up -d --build

python3 benchmark_transcribe.py \
    --url http://localhost:8357 \
    --levels 1,2,4 --num-speakers 4 \
```

```
--out paper/measurements/repetitions/thread_short_r1.csv
```

A.3 Process-replica scaling (Table 3, Figures 2–3)

Direct client-side round-robin across the replica URLs isolates per-replica scaling (it bypasses the gateway).

```
# W-short (8-min AMI, 4 speakers), 1..3 replicas via the sweep
python3 benchmark_transcribe.py \
  --urls http://localhost:8360,http://localhost:8361,http://localhost:8362 \
  --audio testing_audio/dataset/ami/ES2005a.wav \
  --levels 1,2,3 --num-speakers 4 \
  --out paper/measurements/repetitions/process_short_r1.csv

# W-long (40-min Oyez, 8 speakers), three replicas
python3 benchmark_transcribe.py \
  --urls http://localhost:8360,http://localhost:8361,http://localhost:8362 \
  --audio testing_audio/dataset/oyez/2000.99-1978-t01.wav \
  --levels 1,3 --num-speakers 8 \
  --out paper/measurements/repetitions/process_long_3rep_r1.csv

# W-long, four replicas (REPLICAS=4 python3 whisperx_cluster.py up first)
python3 benchmark_transcribe.py \
  --urls http://localhost:8360,http://localhost:8361,\
http://localhost:8362,http://localhost:8363 \
  --audio testing_audio/dataset/oyez/2000.99-1978-t01.wav \
  --levels 4 --num-speakers 8 \
  --out paper/measurements/repetitions/process_long_4rep_r1.csv
```

A.4 Gateway comparison and batch test (Tables 4–5)

Point the load generator at the single gateway URL; nginx distributes submissions and routes status polls back to the owning replica by the job-id prefix. The gateway rows of Table 4 are the `--url` runs at concurrency 3; the direct rows are the `--urls` runs above at the matching concurrency.

```
# W-short gateway comparison (repeat five times)
python3 benchmark_transcribe.py \
  --url http://localhost:8357 \
  --levels 3 --num-speakers 4 \
  --out paper/measurements/repetitions/gateway_short_r1.csv

# W-long gateway comparison (repeat five times)
python3 benchmark_transcribe.py \
  --url http://localhost:8357 \
  --audio testing_audio/dataset/oyez/2000.99-1978-t01.wav \
  --levels 3 --num-speakers 8 \
  --out paper/measurements/repetitions/gateway_long_r1.csv

# Ten-file batches (repeat each three times)
python3 benchmark_transcribe.py \
  --url http://localhost:8357 \
  --audio testing_audio/dataset/ami/ES2005a.wav \
  --levels 10 --num-speakers 4 \
  --out paper/measurements/repetitions/batch_short_r1.csv

python3 benchmark_transcribe.py \
  --url http://localhost:8357 \
  --audio testing_audio/dataset/oyez/2000.99-1978-t01.wav \
```

```
--levels 10 --num-speakers 8 \  
--out paper/measurements/repetitions/batch_long_r1.csv
```

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